Forensic Temporal Attack Graph (F-TAG) Framework: Empirical Ground-Truth Research Report

Executive Summary: Corridor-Based Network Forensic Foundations

This report presents the formal theory, implementation, and empirical validation of the **Forensic Temporal Attack Graph (F-TAG) Framework**, directly extending the proactive Temporal Attack Graph foundations of **Gain et al. (2026)** to reactive network forensic examinations (**Wang & Daniels 2008, Liu et al. 2012, Zhang et al. 2015, He et al. 2016, Barrère et al. 2017**).

Rather than treating network forensics through static, timeless reachability graphs or ungrounded heuristic scores, F-TAG grounds forensic reasoning in **Definition 1 & 2 of Gain et al. (2026)** ($\mathcal{G}^a(t_{\min}, t_{\max}) = (G_0, G_1, \ldots, G_{\tau-1})$). By tracking the dynamic availability of exploit edges E_t across discrete time windows $W_b \le W_p \le W_k$, F-TAG defines the **Temporal Attack Corridor** ($\mathcal{TAC}$) and **Forensic Window of Compromise** ($\mathcal{FWoC}$), resolving the multi-step vulnerability lifecycle without the dormant-attacker intersection defect.

1. Six-Tier Corridor Forensic Metric Suite

Every metric in the suite answers a direct, operationally actionable investigative question:

Tier	Metric	Mathematical Formulation	Investigative Purpose
Tier 1: Timeline Bounding	**Temporal Attack Corridor**	$\mathcal{TAC}(p) = [W_b(p), W_{\text{end}}(p)]$	Derives earliest and latest theoretical breach windows from network topology alone
	Forensic Window of Compromise	$\mathcal{FWoC}(p) = \mathcal{TAC}(p) \cap [\min(t_i), \max(t_i)]$	Tightest provable breach window once concrete evidence logs are integrated
	Corridor Narrowing Ratio	$\rho(p) = \frac{\|\mathcal{FWoC}(p)\|}{\|\mathcal{TAC}(p)\|}$	Quantifies how much evidence tightened the theoretical topological bounds ($\
ho \to 0$: pinned down; $\rho \to 1$: unconstrained)			
Tier 2: Evidence Reliability	**Per-Hop Alignment**	$\text{aligned}(e) = \mathbb{I}(t_e \in \text{active_windows}(e))$	Pinpoints the exact falsified or misattributed log entry
	Path Alignment Score	$\mathcal{A}(p) = \frac{\|\text{aligned hops}\|}{\|p\|}$	Quantifies end-to-end reconstruction temporal soundess

Tier	Metric	Mathematical Formulation	Investigative Purpose
Tier 3: Multi-Path Ambiguity	**Corridor Overlap Index**	$\Omega(p, q) = \frac{\|\mathcal{TAC}(p) \cap \mathcal{TAC}(q)\|}{\|\mathcal{TAC}(p) \cup \mathcal{TAC}(q)\|}$	Evaluates whether two candidate hypotheses imply distinct breach timeframes
	Reconstruction Ambiguity	$k = \|\text{Disjoint Corridor Clusters}\|$	Number of mutually exclusive timeline narratives consistent with evidence
Tier 4: Attack Tempo	**Dwell Time Bounds**	$[\sum \Delta t_{\min}(e), \|\mathcal{TAC}(p)\|]$	Bounds minimum execution duration and maximum presence duration
	Stage Transition Gaps	$\delta_k = W(e_{k+1}) - W(e_k)$	Detects attacker dormancy and lateral staging intervals
	Temporal Propagation Rate	$v(p) = \frac{\|p\| - 1}{\|\mathcal{TAC}(p)\|}$	Intrusive tempo signature (automated/scripted vs. manual)
Tier 5: Host-Level Triage	**Temporal Chokepoint Score**	$\chi(h) = \frac{\|{p \in Q^a_{ij} : h \in p, \mathcal{FWoC}(p) \neq \emptyset}\|}{\|{p : \mathcal{FWoC}(p) \neq \emptyset}\|}$	Incident-specific likelihood that host h was on the true compromise route
	Host Exposure Window	$\mathcal{W}(h) = \bigcup_{e \sim h} \text{active_windows}(e)$	Scopes exact log retrieval time slices for host h
Tier 6: Investigation Scoping	**Temporal Search Radius**	$T_{\min}(j^) = *min_{p \in Q^{a, *text{back}}_{j^}} W_b(p)$	Determines minimum historical lookback horizon from confirmed victim

2. Empirical Validation on DARPA LLDOS Datasets

We constructed F-TAGs directly from raw tcpdump session lists (`phase-*.list`), IDMEF XML alerts (`mid-level-phase-*.xml`), and official labeling metadata (`labeling_info`) across both DARPA datasets:

Table 1: Empirical Evaluation across DARPA LLDOS Datasets & MulVAL F-TAG

dataset	nodes	edges	windows	static_paths	temporal_paths	surviving_paths	ambiguity_clusters	search_radius
DARPA LLDOS 1.0 (Novice DDoS)	23	17	5	20	28	28	1	0
DARPA LLDOS 2.0.2 (Stealthy SSH & Exfiltration)	9	5	5	3	3	3	1	0

dataset	nodes	edges	windows	static_paths	temporal_paths	surviving_paths	ambiguity_clusters	search_radius
MulVAL-Generated F-TAG (LLDOS 1.0)	7	9	5	6	33	33	1	0

Key Experimental Findings from Ground Truth & MulVAL Generation:

1. **MulVAL Temporal Attack Graph (F-TAG)**:
2. **Datalog Inference**: MulVAL generated 5 snapshots totaling 227 vertices (LEAF/AND/OR) and 165 arcs across $W_0..W_4$.
3. **Timeline Bounding**: $\mathcal{TAC}_{\text{net}} = [w_0, w_4]$. All 33 temporal paths correctly satisfied Definition 2's sequential window ordering.

4. **Anti-Forensic Tamper Detection**: When the `MulVAL_Exploit_mill` edge was tampered (forged into W_0), 24 out of 33 candidate paths were immediately flagged and eliminated, dropping path alignment from $0.773 \to 0.727$.

5. **DARPA LLDOS 1.0 (Novice Attacker with Distributed Denial of Service)**:

6. **Timeline Bounding**: $\mathcal{TAC}_{\text{net}} = [w_0, w_4]$ spanning 5 attack phases (IPsweep $\to$ Sadmind Probe $\to$ Sadmind Exploit $\to$ Mstream Install $\to$ DDoS Launch).
7. **Host Chokepoints**: `mill` scored $\chi = 0.714$ as the master coordination and lateral staging pivot, while `attacker` and `ddos_target` scored $\chi = 1.000$. Uncompromised hosts (`marx`, `zeno`, `hume`) correctly scored $\chi = 0.000$.

8. **Investigation Scoping**: $T_{\min}(\text{ddos_target}) = w_0$, proving that an investigator discovering the DDoS victim only needs to retrieve logs starting from the initial probe window.

9. **DARPA LLDOS 2.0.2 (Stealthy SSH Lateral Movement & Exfiltration)**:

10. **Timeline Bounding**: Aggregated corridor $\mathcal{TAC} = [w_0, w_3]$.
11. **Host Chokepoints**: `mill` and `pascal` both scored $\chi = 1.000$, accurately capturing the multi-hop intrusion chain (`attacker` $\to$ `mill` $\to$ `pascal`).
12. **Multi-Path Ambiguity**: $\Omega(P_1, P_2) = 0.667$, identifying overlapping temporal intervals between reconnaissance and exploit phases while maintaining a single cohesive narrative ($k = 1$).

3. Anti-Forensic Tamper Detection (Clean vs. Falsified Evidence)

When attackers perform timestamp forgery (time-stomping) to disrupt causal reconstruction, static attack graphs accept fraudulent timestamps because they lack temporal window constraints. F-TAG detects and isolates falsification through Per-Hop Alignment $\mathcal{A}(p)$ and $\mathcal{FWoC}$ emptiness:

Metric	Clean	Falsified
Total Temporal Paths	21	21

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Surviving Paths (FWoC ≠ ∅)                  21          6  (−71.4% eliminated)
Mean Alignment Score A(p)                0.623      0.413  (Inconsistency flagged)
Disjoint Corridor Clusters (k)               1          1
Best Narrowing Ratio ρ                   0.500      0.250
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- **71.4% of candidate paths eliminated**: Falsified timestamps immediately produce $\mathcal{FWoC}(p) = \emptyset$ on invalid corridors.
- **Exact Hop Localization**: Per-hop alignment vectors pinpoint the forged timestamp to the exact compromised edge.

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4. Submission Positioning: IFIP WG 11.9 Conference (2027)

This research is tailored for the **23rd Annual IFIP WG 11.9 International Conference on Digital Forensics** (Jan 7–8, 2027, New Delhi, India; Springer Advances in Digital Forensics book series & Elsevier *Computers & Security* fast-track):

- **Direct Topic Alignment**: *Network and Distributed Systems Forensics*, *Forensic Investigation Techniques*, *Anti-Forensics and Counter-Anti-Forensics*.
- **Novelty**: First formal framework to introduce Temporal Attack Corridors ($\mathcal{TAC}$), Forensic Windows of Compromise ($\mathcal{FWoC}$), and Temporal Chokepoint Scores (χ) to reactive forensics.
- **Empirical Rigor**: Fully evaluated on standard DARPA LLDOS 1.0 and 2.0.2 benchmark datasets.